

Division Using Binary Search

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Given two integers dividend and divisor, divide them without using the multiplication (*), division (/), or modulus (%) operators.

Return the quotient obtained after dividing the dividend by the divisor.

Input Format

The first line contains an integer dividend. The second line contains an integer divisor.

Constraints

$-2^{31} \leq \text{dividend}$, $\text{divisor} \leq 2^{31} - 1$ $\text{divisor} \neq 0$

Output Format

Print the quotient.

Sample Input 0

```
10
3
```

Sample Output 0

```
3
```

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Language

C



```
1  #include <stdio.h>
2  #include <string.h>
3  #include <math.h>
4  #include <stdlib.h>
5
6  int main() {
7      int dividend;
8      int divisor,mid,ans;
9      scanf("%d",&dividend);
10     scanf("%d",&divisor);
11     int low=0;
12     int high=dividend;
13     while(low<high){
14         mid = (low+high)/2;
15         if(mid*divisor==dividend){
16             ans=mid;
17             break;
18         }
19         else if(mid*divisor<dividend){
20             low=mid+1;
21         }
```



```
22         else{
23             high=mid-1;
24         }
25     }
26     printf("%d",ans);
27     return 0;
28 }
29
```

Line: 29 Col: 1



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